

MATH20250 HW3 Observations

Gwen Jacobson

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Problem-Specific Observations

This time, I have solutions to problems for which I identified areas or concepts with explanations worth repeating. As always, also consult the debrief file on Canvas.

LADW 2.7.4. Prove that if $A : X \rightarrow Y$ and V is a subspace of X then $\dim(AV) \leq \text{rank}(A)$. Deduce from here that $\text{rank}(AB) \leq \text{rank}(A)$.

We have vector spaces V , X and Y , where V is a subspace of X . Recall the following definitions:

1. $\text{rank}(A) = \dim(\text{Ran}(A))$.
2. $\text{Ran}(A) = \text{Col}(A) = \{\mathbf{y} \in Y : \mathbf{y} = A\mathbf{x} \text{ for some } \mathbf{x} \in X\}$.
3. $AV = \{\mathbf{y} \in Y : \mathbf{y} = A\mathbf{v} \text{ for some } \mathbf{v} \in V\}$.

We know that, if C is a subspace of vector space D , $\dim(C) \leq \dim(D)$.¹ From the above definitions, we see that for all $\mathbf{c} \in AV$, $\mathbf{c} \in \text{Ran}(A)$, meaning $AV \subseteq \text{Ran}(A)$. Additionally, since AV contains the zero vector, is closed under vector addition, and is closed under scalar multiplication, AV is a subspace of $\text{Ran}(A)$.

As such,

$$\dim(AV) \leq \dim(\text{Ran}(A)) = \text{rank}(A) \Rightarrow \boxed{\dim(AV) \leq \text{rank}(A)}. \quad \square$$

Consider, then, a matrix B that specifies a transformation with X as its codomain. We then have that

$$\text{Ran}(AB) = \{\mathbf{y} \in Y : \mathbf{y} = A\mathbf{x} \text{ for some } \mathbf{x} \in \text{Ran}(B) \subseteq X\}.$$

From here, it is clear that $\text{Ran}(AB) \subseteq \text{Ran}(A)$, giving

$$\dim(\text{Ran}(AB)) \leq \dim(\text{Ran}(A)) \Rightarrow \boxed{\text{rank}(AB) \leq \text{rank}(A)}. \quad \square$$

¹Suppose, for the sake of contradiction, that $m = \dim(C) > n = \dim(D)$. Then, we can find $\mathbf{c}_1, \dots, \mathbf{c}_m \in C$ that are linearly independent in C and $\mathbf{d}_1, \dots, \mathbf{d}_n \in D$ that are linearly independent in D . However, since $C \subseteq D$, $\mathbf{c}_1, \dots, \mathbf{c}_m$ must also be linearly independent in D , which is impossible if $m > n$. Therefore, for vector space D and a subspace C of D , $\dim(C) \leq \dim(D)$.

LADW 2.8.3. Find the change of coordinates matrix that changes the coordinates in the basis $1, 1+t$ in $\mathbb{P}_1$ to the coordinates in the basis $1-t, 2t$.

Recall that $\mathbb{P}_1 = \text{span}(1, t)$, where t is the first-degree indeterminate. As such, $\mathbb{P}_1 \simeq \mathbb{R}^2$, meaning we can express any polynomial in $\mathbb{P}_1$ uniquely as a vector in $\mathbb{R}^2$. Here, I will have

$$1 \mapsto \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad t \mapsto \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Let $\mathcal{S} = \{1, t\}$ be the standard basis for $\mathbb{P}_1$, and let $\mathcal{A} = \{1, 1+t\}$ and $\mathcal{B} = \{1-t, 2t\}$ be the other two bases provided for $\mathbb{P}_1$. Following Treil's notation, we form the following two identity matrices:

$$[I]_{\mathcal{S}\mathcal{A}} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad [I]_{\mathcal{S}\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}.$$

We will take a second to break down the notation behind $[I]_{\mathcal{S}\mathcal{A}}$. This an **identity matrix** ($[I]$) in that it takes a vector written in terms of the basis $\mathcal{A}$ and returns *the same vector* written in terms of the basis $\mathcal{S}$. In fact, note that the columns of $[I]_{\mathcal{S}\mathcal{A}}$ and $[I]_{\mathcal{S}\mathcal{B}}$ are precisely the representations of the respective $\mathcal{A}$ and $\mathcal{B}$ basis vectors in terms of our standard basis $\mathcal{S}$.

The right-to-left orientation of the subscript is helpful for this, as the “input” vector should be written in terms of the “outermost” basis – here, $\mathcal{A}$. Since the change of coordinates to $\mathcal{S}$ is accomplished by left-multiplying the vector (written in terms of $\mathcal{A}$) by $[I]_{\mathcal{S}\mathcal{A}}$, $\mathcal{A}$ is appropriately put to the right in order to match the input vector.

We want to find $[I]_{\mathcal{B}\mathcal{A}}$ – per Equation 8.1 on pg. 70 of LADW, we have

$$[I]_{\mathcal{B}\mathcal{A}} = [I]_{\mathcal{B}\mathcal{S}}[I]_{\mathcal{S}\mathcal{A}} = ([I]_{\mathcal{S}\mathcal{B}})^{-1} [I]_{\mathcal{S}\mathcal{A}}.$$

Here, we glean some more helpful details from the notation. First, note that simply inverting the order of the bases in an identity matrix's subscript corresponds to inverting the matrix itself!

Second, recall the trick to determine (a) the compatibility of matrices for multiplication and (b) the dimension of the product matrix, should multiplication be possible. Consulting the “graphic” below, we see that multiplying an $m \times n$ matrix on the right by an $n \times p$ matrix is possible, since the underlined quantities (close in proximity to each other) match. We further know that the product will be an $m \times p$ matrix, with the outer boxed quantities representing the dimensions of the domain (p , the rightmost) and target space (m , the leftmost):

$$\boxed{m} \times \underline{n} \cdot \underline{n} \times \boxed{p}.$$

Similarly, with the $[I]_{\mathcal{B}\mathcal{S}}[I]_{\mathcal{S}\mathcal{A}}$ multiplication, we have compatibility with the matching innermost $\mathcal{S}$ s and the assurance that the product takes the initial representation in the rightmost $\mathcal{A}$ to the target representation in the leftmost $\mathcal{B}$ (the “balance of indices”, as Treil puts it).

Employing all of this, we have

$$[I]_{\mathcal{B}\mathcal{A}} = \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \Rightarrow \boxed{[I]_{\mathcal{B}\mathcal{A}} = \begin{pmatrix} 1 & 1 \\ \frac{1}{2} & 1 \end{pmatrix}}.$$

LADW 2.8.4. Let T be the linear operator in $\mathbb{F}^2$ defined (in the standard coordinates) by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x + y \\ x - 2y \end{pmatrix}.$$

Find the matrix of T in the standard basis and in the basis

$$(1, 1)^T, \quad (1, 2)^T.$$

Let $\mathcal{S} = \{(1, 0)^T, (0, 1)^T\}$ be the standard basis for $\mathbb{F}^2$, and let $\mathcal{A} = \{(1, 1)^T, (1, 2)^T\}$ be the other basis. Note that we can form the following matrices:

$$[I]_{\mathcal{S}\mathcal{A}} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \quad \text{and} \quad [T]_{\mathcal{S}\mathcal{S}} = \begin{pmatrix} 3 & 1 \\ 1 & -2 \end{pmatrix}.$$

The formation of $[I]_{\mathcal{S}\mathcal{A}}$ is relatively straightforward from what we did on 2.8.3. As for $[T]_{\mathcal{S}\mathcal{S}}$, we simply perform the specified transformation on each of the standard basis vectors to construct each column of the matrix. The resulting matrix is then able to operate on a vector represented in terms of $\mathcal{S}$ and transform it appropriately; that is, the transformation is geared specifically to give all outputs in terms of $\mathcal{S}$.

We still need the transformation in terms of the basis $\mathcal{A}$. Suppose we have a vector in $\mathbb{F}^2$ in the $\mathcal{A}$ basis representation. Left-multiplying it by $[I]_{\mathcal{S}\mathcal{A}}$ would return its representation in terms of $\mathcal{S}$. Then, left-multiplying that $\mathcal{S}$ -representation product by $[T]_{\mathcal{S}\mathcal{S}}$ would appropriately apply the transformation, returning the transformed vector in the $\mathcal{S}$ representation. The final step is to change the coordinates of the $\mathcal{S}$ -represented transformed vector, returning the same vector represented in the $\mathcal{A}$ basis. Such a change is, as we saw earlier, accomplished by left-multiplying $[I]_{\mathcal{A}\mathcal{S}}$, which is given by

$$[I]_{\mathcal{A}\mathcal{S}} = ([I]_{\mathcal{S}\mathcal{A}})^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}.$$

Bringing everything together, we have

$$[T]_{\mathcal{A}\mathcal{A}} = [I]_{\mathcal{A}\mathcal{S}}[T]_{\mathcal{S}\mathcal{S}}[I]_{\mathcal{S}\mathcal{A}},$$

giving us

$$[T]_{\mathcal{A}\mathcal{A}} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 4 & 5 \\ -1 & -3 \end{pmatrix} \Rightarrow [T]_{\mathcal{A}\mathcal{A}} = \begin{pmatrix} 9 & 13 \\ -5 & -8 \end{pmatrix}.$$

LADW 2.8.5. Prove that, if A and B are similar matrices, then $\text{trace}(A) = \text{trace}(B)$.

We have square matrices A and B with A similar to B . So, there exists an invertible matrix Q such that

$$A = Q^{-1}BQ.$$

From here, we have

$$\text{trace}(A) = \text{trace}(Q^{-1}BQ) = \text{trace}((Q^{-1}B)Q).$$

The reason for the use of associativity on the rightmost side has to do with the hint about the relationship between $\text{trace}(XY)$ and $\text{trace}(YX)$. In particular, Theorem 5.1 in LADW (proved in Exercise 1.5.6, but not here) says that commutable matrices X and Y have the property that $\text{trace}(XY) = \text{trace}(YX)$.

The next step in the current exercise is to employ this property, which is where associativity comes in handy. The theorem does not address the relationship between $\text{trace}(XYZ)$ and $\text{trace}(XZY)$ for commutable matrices X , Y , and Z . Therefore, it is best practice to treat – using associativity – $Q^{-1}B$ as the single product matrix it is and commute *that product* with Q . You may also treat BQ as a single product matrix and commute it with Q^{-1} ; you just want to avoid commuting B *on its own*.

So, continuing, we have

$$\text{trace}(A) = \text{trace}((Q^{-1}B)Q) = \text{trace}(QQ^{-1}B) = \text{trace}(IB) = \text{trace}(B) \Rightarrow \boxed{\text{trace}(A) = \text{trace}(B)}.$$

Supp. V. Give an example of system of linear equations that has exactly 25 solutions, or explain why this is not possible.

Note how this problem does not specify the space(s) in which you must operate – as such, **do not limit yourself to $\mathbb{R}^n$** or the like.

Many answers to this question pointed out the fact that there are 0, 1, or “*infinitely many*” solutions to any system of linear equations. This is only somewhat correct, as “infinitely many” is a bit more subjective than absolute.

For example, the one-equation system $x + y = 0$ in two unknowns has infinitely many solutions – particularly $\text{span}(\{(-1, 1)^T\})$ – if the target space is $\mathbb{R}^2$.

However, consider the same system but with the target space as $(\mathbb{Z}/5\mathbb{Z})^2$. The solution set is still $\text{span}(\{(-1, 1)^T\})$, though we might represent it as $\text{span}(\{(4, 1)^T\})$. However, the cardinality (size) of this set is

$$\left| \text{span} \left(\left\{ \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\} \right) \right| = \left| \left\{ \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} \right| = 5.$$

So, the trichotomy of “0, 1, or infinitely many” needs to be adapted in cases where the target space is finite.

Back to the problem at hand, consider the homogeneous equation $A\mathbf{x} = \mathbf{0}$, where A is the zero matrix in $M_{2 \times 2}(\mathbb{Z}/5\mathbb{Z})$ and $\mathbf{x} \in (\mathbb{Z}/5\mathbb{Z})^2$. Every vector in $(\mathbb{Z}/5\mathbb{Z})^2$ is a solution, meaning the cardinality of the solution set is $|(\mathbb{Z}/5\mathbb{Z})^2| = 5^2 = 25$.

General Observations

IMPORTANT: Any problems with this boxed “x” next to them – **X** – should NOT be chosen for your HW submission. If you do this on one HW assignment, you will be warned but given credit. On subsequent assignments, you will not receive credit for problems that have been so marked.

- **Observations on Proofs**

- (a) **Directionality:** Suppose you have to prove “ A if and only if B .” ($A \Leftrightarrow B$) This is a **twofold** process:
 - i. **Prove “if A , then B .”** This may be done by supposing A is true and proving B directly, supposing A is true but B is false and proving by contradiction, or by contrapositive – “if A , then B ” is logically equivalent to “if not B , then not A .”
 - ii. **Prove “if B , then A .”** This may be done by supposing B is true and proving A directly, supposing B is true but A is false and proving by contradiction, or by contrapositive – “if B , then A ” is logically equivalent to “if not A , then not B .”
- (b) **Defining variables and other quantities:** Whenever you introduce a new quantity, be it a scalar, vector, etc., **state the space in which it lives.**

Ex. $\mathbf{v} \in V$ (where V is a vector space), $\alpha \in \mathbb{C}$.

The $\in$ symbol, meaning “is an element of,” is generated by `\in`.

- (c) **Existence:** If you are tasked with proving the existence of a quantity, **do not presuppose its existence in your notation.**
 - i. Example: LADW 1.6.7 – The goal is to prove that the matrix B is invertible given that the matrices A and AB are invertible. It may be tempting at some point in the proof to use the following fact from Theorem 6.3 (pg. 26):

$$(AB)^{-1} = B^{-1}A^{-1}.$$

However, there is a key condition attached to that fact:

If linear transformations A and B are **invertible**...

As such, this fact should not be employed unless you have already shown that A and B are invertible.

- (d) **Justifications:** There are two main things to remember when justifying the steps in a proof:
 - i. **Prove all facts on which you rely.** If you need to use a result that you did not already prove in class, **prove it explicitly.** If you need to use a result that comes from another homework problem, **prove it explicitly.** Do not take other homework problems for granted.
 - ii. If you are unsure whether to justify something explicitly, **do it.** Better safe than sorry.

- Miscellaneous

- (a) **Check your calculations.** Especially beware of dropping negative signs and accidentally performing the wrong arithmetic operation on two numbers (e.g., adding instead of multiplying).
 - i. If you are finding a solution to a linear system of equations, **substitute your answer to check**. Sometimes, after converting to matrix form, it can be difficult to remember that there was an underlying group of equations. Use them to your advantage.
 - ii. If you are decomposing a matrix (e.g., writing it as a product of elementary matrices), **multiply the resulting matrices** to make sure you get the result you expect.
- (b) **Read the problems carefully.** Make sure you answer **every part of the problem**.
 - i. This also means you should not make any additional assumptions than what the problem allows. For example, on Supp. V, there was no information about the space(s) in which the equations and solutions had to live, so no such assumptions should have been made.
 - ii. Additionally, for true/false questions, if a statement is true under a certain set of circumstances but *not under the set of circumstances outlined* in the statement, **the statement is false**. If no more specifics are given, err on the side of generality.